



Mahindra University Hyderabad
École Centrale School of Engineering
Minor-I exam

Program: B. Tech. Branch: CIE, ECM, ICS, CSE, AI, CM, NT, COM Year: II
Semester: II
Subject: Numerical Methods (MA2208)

Date: 23/02/2026
Time Duration: 1.5 Hours

Start Time: 10:00 AM
Max. Marks: 15

Instructions:

- 1) Answer all the questions.
- 2) All questions are self-explanatory; no clarification will be provided during the exam.
- 3) Use of non-programmable scientific calculator is allowed. However, sharing calculators during exams is strictly prohibited.

Question 1 (3 marks)

A real number is stored in a computer using the IEEE-754 single-precision (32-bit) floating-point format as

0 10000001 110000000000000000000000

Identify the number in decimal form.

Question 2 (4 marks)

Using Newton-Raphson method perform two iterations to find a root of $x^3 - 7x^2 + 8x - 3 = 0$, with $x_0 = 5$.

Question 3 (4 marks)

Find the order of convergence of following iterative method to determine a root of $f(x) = x^2 - a$, $a > 0$, $x_0 > 0$,

$$x_{n+1} = \frac{x_n}{2} \left(3 - \frac{x_n^2}{a} \right), \quad n = 0, 1, 2, \dots$$

Question 4 (4 marks)

Solve the following system of equations using Gauss elimination method

$$\begin{aligned} x + y + z &= 3, \\ 2x - y - z &= 3, \\ x - y + z &= 9. \end{aligned}$$